Code No: 113AU **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, October - 2020
ELECTRONIC DEVICES AND CIRCUITS
(Common to EEE, ECE, CSE, EIE, IT, MCT)

Time: 2 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the operation of PN Diode in forward and reverse bias with neat diagram.
b) Derive an expression for diode current equation. [8+7]

2.a) Explain the operation and V-I characteristics of Zener diode.
b) Explain the operation and characteristics of SCR. [8+7]

3.a) Explain the operation of Full wave rectifier and derive the expression for efficiency of rectifier circuit.
b) A 230 V, 50HZ voltage is applied to the primary of 4:1 step down transformer used in a Bridge rectifier having a load resistance of 600Ω. Assume the diodes to be ideal. Determine i) DC output voltage ii) DC power delivered to the load iii) PIV and iv) output frequency. [7+8]

4.a) Derive the expression for ripple factor of Half wave Rectifier with C filter.
b) Compare Half wave and full wave and bridge rectifiers. [8+7]

5.a) Explain the operation of transistor in CB configuration.
b) Explain input and output characteristics of a transistor in CB configuration. [7+8]

6.a) Draw CE hybrid model of CE transistor configuration.
b) Explain how Transistor acts as an Amplifier. [8+7]

7.a) Explain the need of biasing.
b) In a self bias circuit V_{cc}=22.5 V, R_c= 5.6 K Ω , R_2= 10 KΩ, R_1= 90 KΩ, h_{fe}=55, V_{BE}=0.6 V. The transistor operates in active region. Determine i) operating point and ii) stability factor. [7+8]

8.a) Explain the operation of p- channel JFET with the help of diagram.
b) Explain pinch off voltage and drain characteristics of JFET. [8+7]

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